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brca12-wgs: a reproducible Nextflow pipeline for *BRCA1/BRCA2* germline analysis from genome, exome and panel sequencing on GRCh38

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Software: https://github.com/pjlabsdev/brca12-wgs · DOI: 10.5281/zenodo.24173638 (version 1.0.1) · Licence: MIT

Summary. Germline pathogenic variants in *BRCA1* and *BRCA2* cause hereditary breast and ovarian cancer syndrome. The relevant spectrum spans single-nucleotide variants, small insertions and deletions, exon-level copy-number changes and larger rearrangements, yet many open BRCA workflows target amplicon panels on GRCh37 and report small variants only. *brca12-wgs* is a Nextflow pipeline that takes FASTQ files or a BAM/CRAM from whole-genome, exome, hybrid-capture panel or amplicon sequencing and produces a structured BRCA1/BRCA2 report on GRCh38. It combines BWA-MEM, GATK HaplotypeCaller or DeepVariant, Manta, CNVkit with per-exon depth ratios, Ensembl VEP with MANE Select transcripts, ClinVar and mosdepth, and lists automated hints on single ACMG/AMP criteria following the ClinGen ENIGMA specification without assigning a classification. We measured performance on four kinds of whole-genome data: Genome in a Bottle truth sets (364 of 364 SNVs and 67 of 69 indels), ten 1000 Genomes carriers of pathogenic variants (10 of 10 detected), 185 small and 12 large variants spiked into real reads, and real Alu insertions and deletions in 1000 Genomes samples, checked in long reads. On 39 samples with assembly-based truth from the Human Pangenome Reference Consortium it found 590 of 590 exonic SNVs without a false positive, and on two exomes and an amplicon panel 33 of 37 SNVs inside the targets. Spiked variants at 20 % allele fraction were all reported, with mosaic candidates from Mutect2 below the germline caller's range. Real Alu insertions were called in only 4 of 11 carriers, but an insertion signature from clipped read ends, added for this reason, was present in 11 of 11. Manta reported all 12 spiked large events, whereas single-sample CNVkit segmentation reported 3 of 12 after a correction made during this work (0 before). The software is for research use; results require confirmation by a clinically validated method.

Availability. Source code, test data, validation scripts and results are available under the MIT licence at https://github.com/pjlabsdev/brca12-wgs.

Highlights

- One command from WGS reads to a BRCA1/BRCA2 report with small variants, structural variants, copy number and exon coverage on GRCh38.
- 10 of 10 real carriers of pathogenic variants from the 1000 Genomes Project were detected with the correct genotype.
- 185 of 185 spiked SNVs and indels up to 40 bp and 12 of 12 spiked deletions, duplications and an Alu insertion were reported.
- Real Alu insertions were called in 4 of 11 carriers; a read-end signature added for this reason was present in 11 of 11 and in 0 of 47 non-carrier evaluations.
- Against assembly-based truth in 39 HPRC samples: 590 of 590 SNVs in the exons ±20 bp, no false positive.
- Exome and amplicon data: 33 of 37 SNVs inside the targets; the misses lie in a UTR stretch without reads.
- Mosaic candidates (Mutect2) and the germline caller together reported every spiked variant at 20 % allele fraction, with no false positive.
- The validation exposed and fixed three defects, including missed copy-number segments.
- Research use only. No classification is assigned.

Research use only. brca12-wgs is not a medical device and has not been clinically validated. Its output must not be used for diagnosis, risk assessment or treatment decisions without confirmation in an accredited laboratory (for example Sanger sequencing for small variants and MLPA for copy-number changes). ACMG/AMP entries are automated hints on individual criteria, not variant classifications. A result without findings does not exclude a hereditary BRCA1/BRCA2 predisposition.

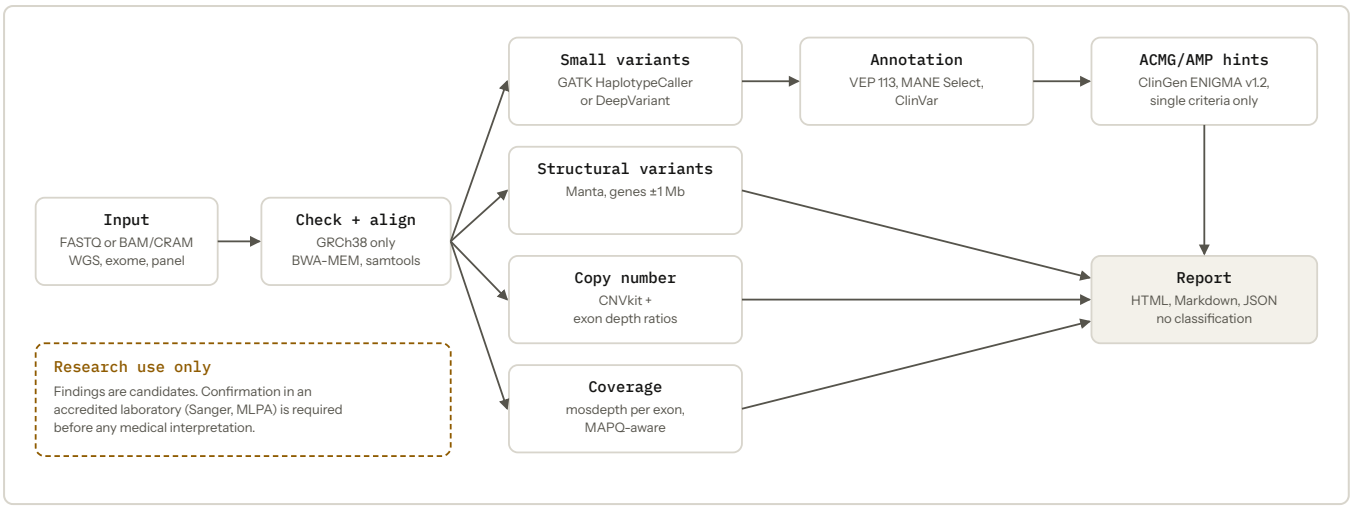


Figure 1. Workflow. Reads are checked against GRCh38 and aligned. Four evidence layers are computed independently and kept separate in the report. Hints on single ACMG/AMP criteria are attached to annotated small variants; no combination rule is applied.

1 Introduction

Pathogenic germline variants in the tumour suppressor genes *BRCA1* (chr17, minus strand) and *BRCA2* (chr13, plus strand) substantially increase lifetime risks of breast, ovarian, prostate and pancreatic cancer [1]. Most pathogenic alleles are truncating SNVs and small indels, but large genomic rearrangements such as single- or multi-exon deletions and duplications account for a relevant fraction of findings, particularly in *BRCA1*, which is rich in Alu repeats and has a partially duplicated pseudogene (*BRCA1P1*) near its 5′ end [2]. Insertions of mobile elements are rare but include a founder variant, the Alu insertion c.156_157insAlu in *BRCA2* exon 3 [3, 4].

Whole-genome sequencing (WGS) captures all of these variant classes in a single assay, but extracting a reliable gene-level result requires several specialised tools, consistent coordinates and explicit statements about where the data are not sufficient. Openly available workflows such as BRCA-analyzer [5] were developed for amplicon panels on GRCh37/hg19, depend on licence-restricted annotation and do not report copy-number or structural variants. General-purpose germline pipelines such as Sarek [6] cover WGS but are not tailored to a concise, gene-focused report with exon-level coverage and conservative interpretation hints. Because many laboratories sequence exomes or gene panels rather than genomes, the same report should also be available for targeted data.

brca12-wgs addresses this gap with three design goals: (i) GRCh38 only, with automatic rejection of other assemblies; (ii) separate, traceable evidence layers for small variants, copy-number candidates, structural variants and coverage; and (iii) reproducibility through pinned software environments, recorded tool versions and a verbatim command log.

2 Implementation

The pipeline is written in Nextflow DSL2 [7] and runs locally with conda environments, a Docker or Singularity image, or tools on the system path (Figure 1, Table 1).

Table 1. Processes, tools and outputs.

Step	Tool (pinned version)	Purpose
Input check	Python	GRCh38 detection from chr13/chr17 lengths; region and exon BED harmonisation
Alignment	BWA-MEM 0.7.18 [8], samtools 1.21 [9]	Alignment, fixmate, sort, duplicate marking (not for amplicons), primer clipping with `ampliconclip`
Read QC	samtools stats, mosdepth	Mapping, pairing, duplicates, error rate; per target depth, uniformity, fold-80, on-target reads
SNVs/indels	GATK 4.6.1.0 HaplotypeCaller [10] or DeepVariant [11]	Calling in both genes ± 200 bp
Mosaic candidates	GATK Mutect2 [12] (optional)	Calls at 2–30 % allele fraction not made by the germline caller
Structural variants	Manta 1.6.0 [13]	Breakpoints within ± 1 Mb of both genes
Copy number	CNVkit 0.9.11 [14]	WGS, hybrid or amplicon mode, flat or pooled reference; per-exon depth ratios
Annotation	Ensembl VEP 113 [15], bcftools	MANE Select [16], HGVS, gnomAD [17], ClinVar [18]
Coverage	mosdepth 0.3.10 [19]	Per exon at MAPQ ≥ 20 and ≥ 0; 10/20/30×
Report	Python (standard library)	HTML, Markdown, PDF, Excel (English, German or Russian), JSON

2.1 Reference and regions

The input check reads the FASTA index and accepts the reference only if chr13 and chr17 have the GRCh38 lengths; GRCh37 is rejected with an explicit message. Gene loci are fixed to chr17:43,044,295–43,170,327 (*BRCA1*) and chr13:32,315,077–32,400,268 (*BRCA2*). Exons are taken from the MANE Select transcripts [16] NM_007294.4 (23 exons) and NM_000059.4 (27 exons). For *BRCA1* the report also shows the legacy exon numbering, which omits exon 4.

2.2 Variant calling

Reads are aligned with BWA-MEM [8]; sorting, mate fixing and duplicate marking use samtools [9]. Small variants are called with GATK HaplotypeCaller [10] restricted to the padded gene regions, or optionally with DeepVariant [11]. Manta [13] is run with call regions extending 1 Mb around both genes, so that rearrangements with one breakpoint inside a gene are retained, including breakend (BND) pairs. Two breakends a few bases apart whose mates lie elsewhere are flagged as a possible insertion of foreign sequence such as a mobile element. In addition, soft-clipped read ends inside the genes are searched for the signature of an insertion: reads from the left clipped at one position, reads from the right clipped at a position up to 30 bp before it (the target-site duplication), at least three clipped reads on each side and a clipped fraction of at least 15 %. Such a pair is listed as an insertion signature when the clipped sequence contains k-mers of the Alu consensus; it is a hint from the reads, not a variant call. A second signature marks an extra copy of a tandem repeat unit: reads clipped at the end of a unit whose clipped part matches the start of that unit in the reference, and reads clipped at its start whose clipped part matches its end, both pointing to the same unit length of 50 to 1,000 bp, with at least two reads at each border (clip positions within 10 bp pooled). For the common extra copy upstream of *BRCA1* exon 1 a marker base is checked as well: the extra copy repeats unit 1, which differs from unit 2 at chr17:43,128,365, so its reads placed on unit 2 show C instead of T there. A third signal is the haplotype the copy sits on: five SNPs 3 to 21 kb downstream that are inherited together with it in the assemblies and long reads (Section 3.7). The known copy is reported when two of the three signals (clipped reads, marker base, at least three tag SNPs with the alternative allele) agree, and the number of haplotype copies gives the likely zygosity. Known alleles from the assemblies and long reads (this copy, the common intronic *BRCA2* Alu insertion and a 980 bp *BRCA1* intronic deletion) are in addition genotyped by allele-aware realignment, in the spirit of graph genotypers such as Paragraph [20] and PanGenie [21] but restricted to the two genes: every read near the site, unmapped mates included, is aligned to the reference haplotype and to the haplotype with the allele (edit distance with free ends, bit-vector algorithm [22]); a read supports the haplotype it fits with at least two fewer edits, and two allele-supporting reads make a call. The tandem copy is reported on this evidence alone or on two of the three other signals. For copy number, CNVkit [14] is run in WGS mode with 1 kb bins over chr13 and chr17 and a flat reference; bins with almost no coverage are excluded from segmentation (Section 3.3). Segments overlapping a gene with an estimated copy number different from two are reported as candidates with the affected exons. Independently, the mean depth of each exon is divided by the median depth of 100 kb windows on the same chromosome; ratios below 0.65 or above 1.35 are listed as dosage hints. Because only a single sample is available, both are labelled as candidates that require MLPA [23] or another validated method.

2.3 Annotation and ACMG/AMP hints

VEP [15] annotates consequences, HGVS notation and exon or intron number on the MANE Select transcript. ClinVar [18] significance, review status and condition come from a user-supplied ClinVar VCF, so that the ClinVar version is under the user's control. The pipeline then evaluates single ACMG/AMP criteria [24] following version 1.2 of the ClinGen ENIGMA BRCA1/BRCA2 specification [25]: BA1, BS1 and BS1_Supporting from gnomAD [17] non-cancer filtering allele frequencies in non-founder populations, PM2_Supporting for substitutions absent from both gnomAD data sets, PP3 and BP4 from BayesDel [26] and SpliceAI [27] inside the defined functional domains, BP1_Strong outside them and BP7 for silent and deep intronic variants. The gnomAD, SpliceAI and BayesDel files are supplied by the user; without them the corresponding hints are reported as open or as approximations. Truncating and canonical splice variants are reported as PVS1 candidates, restricted in the last exon [28]; the exon-specific PVS1 decision tree is not yet applied. No combination rule is used and no classification is assigned; the JSON field `clinicalClassification` is always null.

2.4 Coverage and uncertain regions

Depth per exon is computed with mosdepth [19]. An exon is flagged as uncertain if its mean depth is below 20×, if fewer than 95 % of its bases reach 20×, or if more than 10 % of reads have mapping quality below 20, which identifies regions affected by pseudogene homology or repeats. All intervals below 10× are listed. Small variants receive quality flags for non-PASS filters, low depth, low genotype quality, heterozygous allele fractions outside 25–75 %, low mapping quality and location in an uncertain exon. All thresholds are parameters.

2.5 Reporting and reproducibility

The report has separate sections for *BRCA1* and *BRCA2* small variants, copy-number candidates, structural variants, coverage per exon and uncertain intervals, followed by tool versions, the invocation and limitations. It is pro-

duced as HTML and Markdown in English or German and as JSON (schema `genome.brca12.v1`). Tool versions are captured at run time, every task copies its command script into `brca12_commands.sh`, and conda environments pin all tool versions. Unit tests and a stub run of the complete workflow run in GitHub Actions on every change.

2.6 Targeted sequencing and low allele fractions

With `--assay exome`, `panel` or `amplicon` and a target BED, the target intervals inside both genes are kept and exons without a target are reported as not assessable. Amplicon reads are not marked as duplicates; with a primer BED, primers are removed with `samtools ampliconclip` and the share of reads with a clipped primer is reported. CNVkit runs in hybrid or amplicon mode, optionally with a pooled normal reference, and the exon dosage baseline becomes the median exon depth of the sample. Single-end reads are accepted (Manta is skipped), and `--platform iontorrent` adds a warning to every indel. A read QC step reports mapping, pairing, duplicate and error rates and, for targets, mean depth, uniformity, fold-80 base penalty, on-target reads and weak targets. `--mosaic` adds Mutect2 candidates at low allele fraction (Section 3.6).

3 Validation

Eight data sets were used (Figures 2 and 3). All scripts, sample lists and result files are in the `validation/` directory of the repository.

3.1 Genome in a Bottle truth sets

Small-variant calls were compared with the GIAB NIST v4.2.1 GRCh38 truth sets for HG001 and HG002 [29, 30]. Illumina HiSeq 2×148 bp read pairs that the GIAB alignments placed within either gene ±10 kb were extracted, downsampled to about 36× and 30×, converted back to FASTQ and processed from FASTQ. Calls were matched with RTG Tools `vcfeval` [31], which compares haplotypes and requires the genotype to agree, inside the GIAB high-confidence regions (91.9 % and 95.1 % of the 211,225 bp of both loci). The pipeline recovered 364 of 364 SNVs and 67 of 69 indels, and all 34 variants in the exons ±20 bp without a false positive. The three discordant calls were intronic; two were genotype errors in homopolymers at 12× and 14× that carried the pipeline's own quality flags.

3.2 Real carriers of pathogenic variants

The 30× joint call set of the 3,202 samples of the 1000 Genomes Project [32] was searched for loss-of-function variants in the MANE coding sequence and its canonical splice sites. Ten individuals carry eight such variants, seven of them annotated as pathogenic in the ClinVar release used by that call set (Table 2). Read pairs over both genes ±10 kb were taken from the public CRAM files and processed from FASTQ. All 10 variants were called heterozygous with genotype quality 99, including a 5 bp deletion that the 1000 Genomes call set itself marks as low confidence. As a secondary check, calls in the exons ±20 bp agreed with the 1000 Genomes genotypes of the same individuals for 141 of 141 SNVs and 19 of 21 indels, with 2 and 2 additional calls; that call set is a comparison, not a truth set.

Table 2. Carriers of pathogenic *BRCA1*/*BRCA2* variants in the 1000 Genomes Project and the result of the pipeline. AD, reads supporting reference and variant allele.

Sample	Pop.	Gene	Variant (MANE)	Consequence	Detected, genotype	Depth	AD
HG02164	CDX	*BRCA1*	`c.34C>T`	p.Gln12Ter, nonsense	yes, 0/1	47	23,24
HG03929	BEB	*BRCA1*	`c.3403C>T`	p.Gln1135Ter, nonsense	yes, 0/1	26	14,12
HG00365	FIN	*BRCA1*	`c.4357+1G>A`	canonical splice donor	yes, 0/1	33	16,17
NA19397	LWK	*BRCA2*	`c.2279_2283del`	frameshift	yes, 0/1	25	10,15
HG02274	PEL	*BRCA2*	`c.5146_5149del`	frameshift	yes, 0/1	25	20,5
NA19130	YRI	*BRCA2*	`c.5241_5242insTA`	frameshift	yes, 0/1	36	21,15
NA19132	YRI	*BRCA2*	`c.5241_5242insTA`	frameshift	yes, 0/1	28	9,19
HG01377	CLM	*BRCA2*	`c.7063G>T`	p.Glu2355Ter, nonsense	yes, 0/1	32	20,12
HG00611	CHS	*BRCA2*	`c.7697dup`	frameshift	yes, 0/1	46	27,19
HG00612	CHS	*BRCA2*	`c.7697dup`	frameshift	yes, 0/1	25	20,5

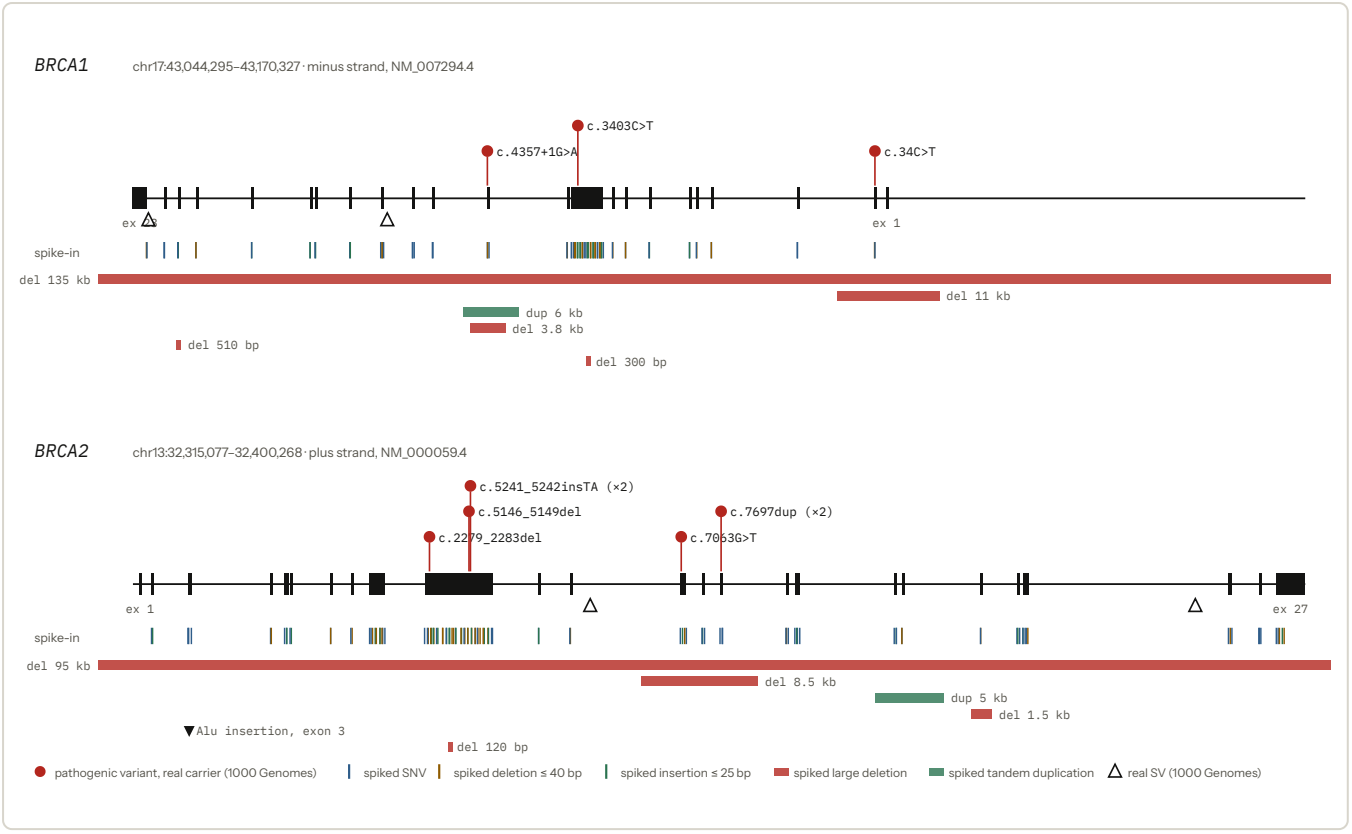


Figure 2. Variants used for validation, drawn on both gene loci. Black boxes are MANE Select exons. Above the gene: pathogenic variants of real carriers. Below: spiked small variants (one tick each), spiked large events and the positions of real structural variants.

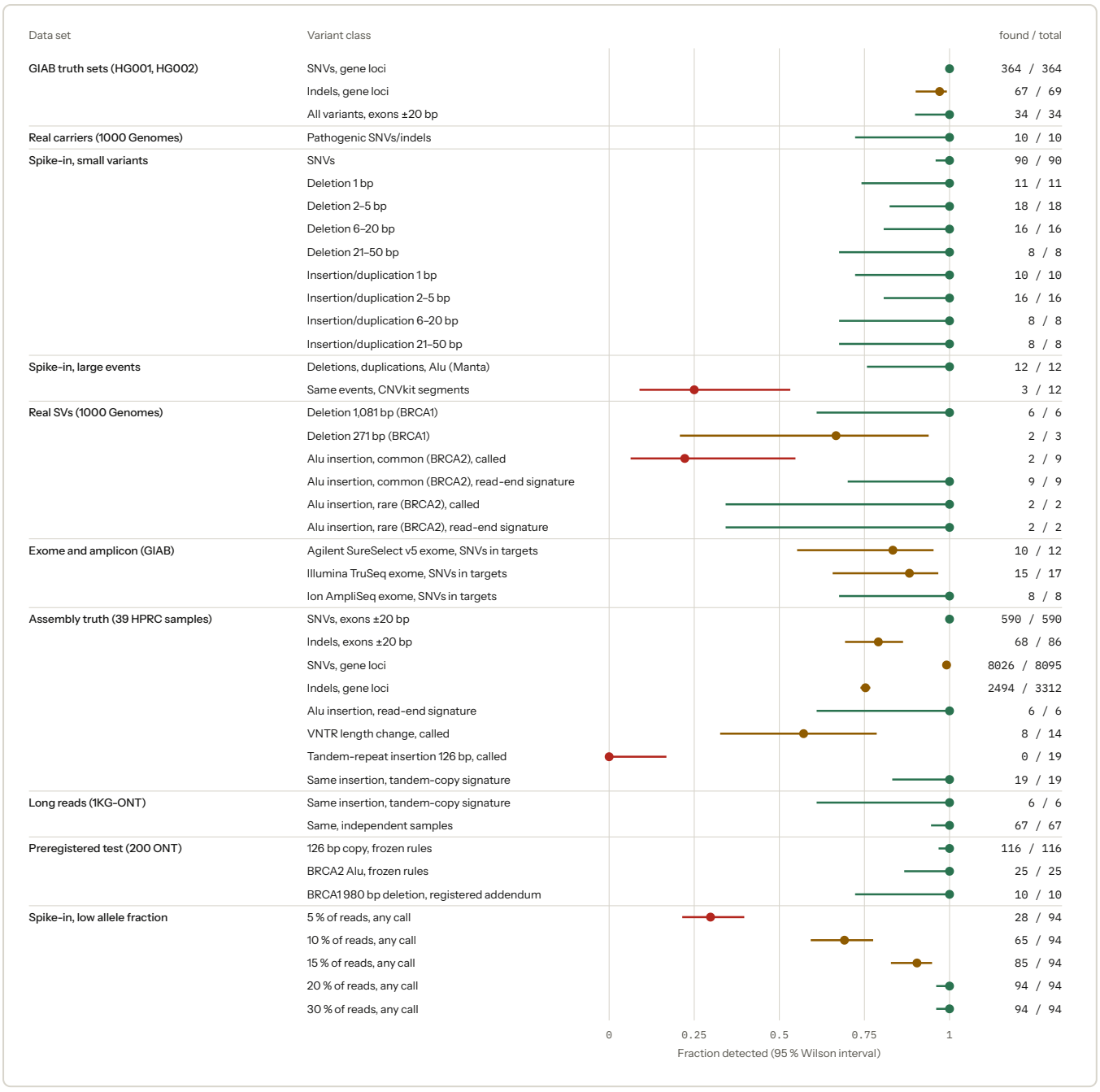


Figure 3. Detection by data set and variant class. Points are the fraction of known variants that the pipeline reported; bars are 95 % Wilson intervals [33]. Green, all found; amber and red, partly found. Small numbers give wide intervals.

3.3 Spike-in experiment

No openly available short-read WGS data set of a carrier of a pathogenic exon deletion, duplication or Alu insertion was found; the GeT-RM reference materials [34] are distributed as DNA. We therefore spiked synthetic heterozygous variants into real reads. The background was GIAB HG002 on chr13 and chr17 at about 30×, which gives CNVkit a real chromosome-wide baseline. Inside both genes ±20 kb every real read pair was replaced by a pair with the same position, fragment length, read lengths and base qualities, read from one of two haplotypes: the reference, or the reference with the spiked events. Sequencing errors were drawn from the real base qualities. Coverage profile and fragment sizes therefore stayed those of the real run. The replaced pairs were aligned with BWA-MEM and merged with the untouched background, and the pipeline was run with real Manta and CNVkit.

Two samples carried 185 small variants: one per coding exon segment with rotating types, SNVs at canonical splice sites, and five variants modelled on well-known pathogenic alleles (*BRCA1* c.68_69del, c.181T>G, c.5266dup; *BRCA2* c.771_775del, c.5946del), whose coordinates were derived from a coding-sequence model that is checked by translation. All 185 were called with the correct genotype, with no false positive in the exons (Figure 3). Outside the exons, every sample including the negative control showed the same two calls at one intronic *BRCA2* site, which come from real HG002 reads with an unmapped mate that the replacement does not cover,

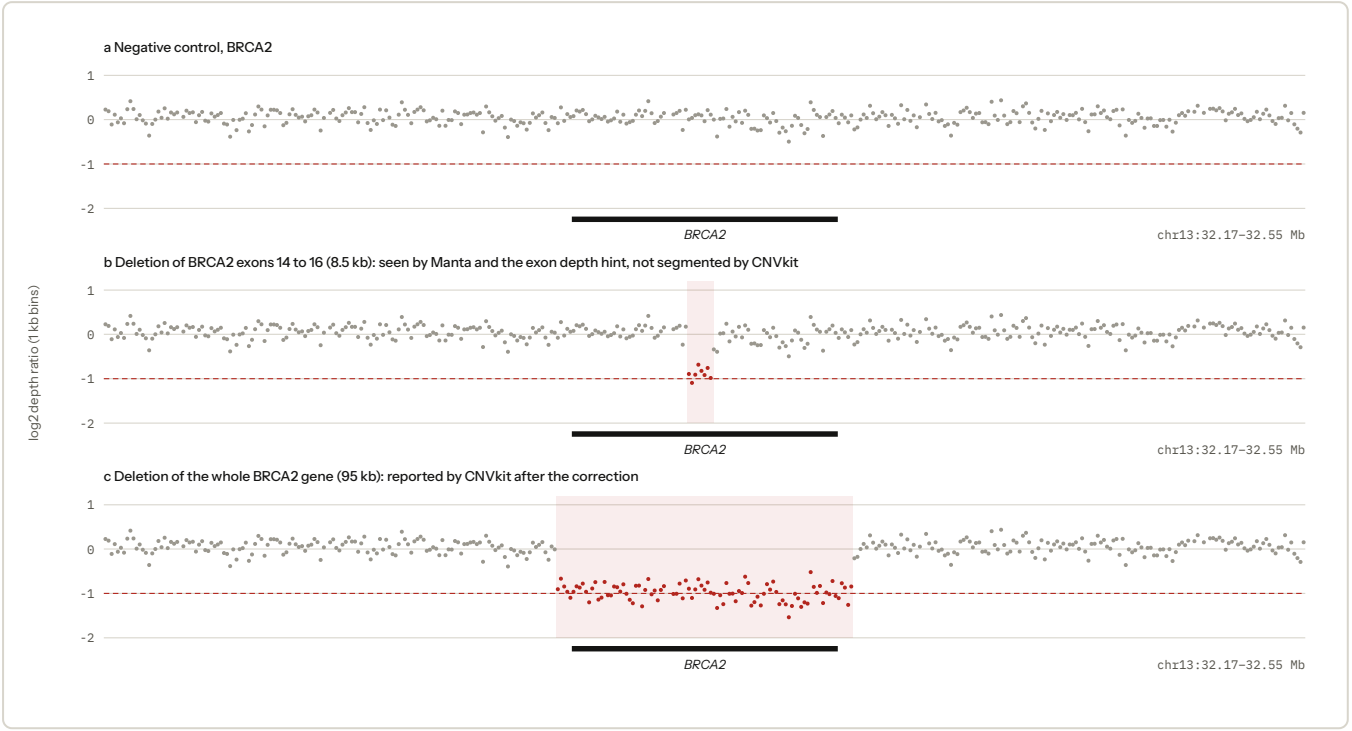
and the samples with large deletions showed short artefactual calls directly at some breakpoints.

Six samples carried 12 large events from 120 bp to a whole-gene deletion, two tandem duplications and an Alu insertion in *BRCA2* exon 3 (Figure 4). Manta reported 12 of 12 with exact breakpoints and heterozygous genotype; the Alu insertion appeared as a pair of breakends. The read-end signature (Section 2.2) listed the spiked Alu insertion and nothing in the other eight spike-in samples. With the settings of version 1.0.2, CNVkit reported 0 of 12: bins without coverage dominated the segmentation statistics, and even whole-gene deletions with a clear half-depth signal were merged into their neighbours (Figure 5). Excluding such bins (`--drop-low-coverage`) raised this to 3 of 12; the smallest event reported by CNVkit was 11 kb. In addition, exon dosage hints had been suppressed for exactly the affected exons, because a heterozygous deletion also makes an exon fall below the depth threshold. After the correction, 9 of the 11 deletions and duplications produced a dosage hint for an affected exon. A third defect stopped the pipeline when the input BAM was named like the sample.

Spiked event (heterozygous)		Size	Manta	CNVkit 1.0.2	CNVkit 1.1.0	Exon depth hint	Small-variant caller
`BRCA2 Deletion`	120 bp deletion inside exon 11	`120 bp`	●	○	○	○	●
`BRCA1 Deletion`	300 bp deletion inside exon 10	`300 bp`	●	○	○	○	○
`BRCA2 Alu insertion`	exon 3 between c.156 and c.157	`327 bp`	●	n/a	n/a	n/a	○
`BRCA1 Deletion`	exon 21	`510 bp`	●	○	○	●	n/a
`BRCA2 Deletion`	exon 21	`1.5 kb`	●	○	○	●	n/a
`BRCA1 Deletion`	exon 12	`3.8 kb`	●	○	○	●	n/a
`BRCA2 Tandem duplication`	exons 19 and 20	`5 kb`	●	○	○	●	n/a
`BRCA1 Tandem duplication`	exon 12	`6 kb`	●	○	○	●	n/a
`BRCA2 Deletion`	exons 14 to 16	`8.5 kb`	●	○	○	●	n/a
`BRCA1 Deletion`	exons 1 and 2 including the promoter	`11 kb`	●	○	●	●	n/a
`BRCA2 Deletion`	the whole gene	`95 kb`	●	○	●	●	n/a
`BRCA1 Deletion`	the whole gene	`135 kb`	●	○	●	●	n/a
Detected			`12 / 12`	`0 / 12`	`3 / 12`	`9 / 12`	`1 / 12`

Filled: reported with the correct type at the spiked position. Open: not reported. Size bar on a log scale.

`Figure 4. Spiked large events and the evidence layer that reported them.` CNVkit 1.0.2 and 1.1.0 differ only in the exclusion of bins without coverage. Single-sample depth methods are not expected to resolve events of one or a few 1 kb bins; n/a marks layers that cannot apply.



`Figure 5. Read-depth signal of spiked heterozygous deletions.` Each point is a 1 kb bin (CNVkit log2 ratio). The dashed line marks the expected value for one copy (–1). Shaded: spiked interval. The signal is present in the bins; whether it is reported depends on the segmentation.

3.4 Real structural variants

The phased SNV, indel and SV panel of the 1000 Genomes Project [32] contains two Alu insertions in *BRCA2* and two deletions in *BRCA1*, all intronic or otherwise not disease-causing. They allow the same kind of event to be tested in real reads (Table 3). Carriers and six samples that are homozygous reference at all four sites were processed as in Section 3.2, with one addition: mates that the original alignment had placed elsewhere in the genome were retrieved as well, because read pairs with a mate inside the inserted element are otherwise lost: without them only 2 of 11 Alu insertions were called and 9 of 11 showed the read-end signature. At the two Alu sites the panel genotypes were looked up for all 33 samples of Sections 3.2 and 3.4, which added three heterozygous carriers of the common insertion; the ten samples of Section 3.2 were rerun with retrieved mates for this purpose. The panel calls are statistically phased short-read calls (the Alu insertions come from MELT [35] within the GATK-SV pipeline [36]) and serve as a comparison, not as truth.

Table 3. Real structural variants in 1000 Genomes samples. Called: by Manta or the small-variant caller at the site. Signature: insertion signature from clipped read ends. Controls: samples with panel genotype 0|0 and any call or signature at the site.

Gene	Event	Position	Panel caller	Called	Signature	Controls
BRCA2	Alu insertion, 281 bp	`chr13:32,348,302`	MELT (GATK-SV)	2 / 9	9 / 9	0 / 22
BRCA2	Alu insertion, 281 bp	`chr13:32,392,272`	MELT (GATK-SV)	2 / 2	2 / 2	0 / 25
BRCA1	Deletion, 271 bp	`chr17:43,045,958`	svtools	2 / 3	n/a	0 / 6
BRCA1	Deletion, 1,081 bp	`chr17:43,071,626`	Manta, Wham (GATK-SV)	6 / 6	n/a	0 / 6

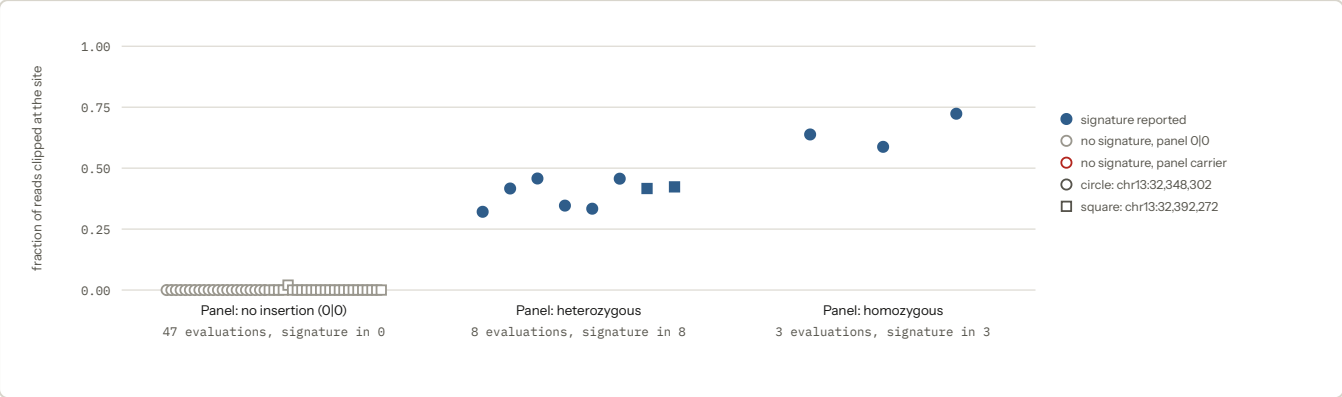


Figure 6. Read ends at the two real Alu insertion sites. Each symbol is one sample at one site: the fraction of reads over the site that are clipped there by at least 10 bases, grouped by the genotype in the 1000 Genomes panel. Filled: the pipeline listed an insertion signature with Alu sequence. The raw fraction counts every read within 15 bp of the site and is therefore lower than the clipped fraction computed by the detector.

The 1,081 bp deletion was reported by Manta in 6 of 6 carriers, with the genotype of the panel including the homozygous carrier; its start lies 101 bp from the panel coordinate, whose breakpoints are not base-exact. The 271 bp deletion was reported in 2 of 3 carriers. In the carrier without a call (HG01796) the depth inside the deleted interval equals the depth next to it (32x and 33x; 13x and 31x in the carrier HG02601), so the reads of this sample do not show the deletion. The Alu insertions were called in 2 of 9 carriers of the common and 2 of 2 carriers of the rare insertion, by Manta as insertions (about 320 bp where the length was resolved) or by the small-variant caller as a partial insertion. The read-end signature was present in 9 of 9 and 2 of 2 carriers, with clipped fractions of 0.43 to 0.64 in heterozygous and 0.92 to 0.94 in homozygous carriers according to the panel. No sample with panel genotype 0|0 had a call or signature at any of the four sites.

3.5 Exome and amplicon data

Three public GIAB data sets from targeted sequencing [37] were run from FASTQ with `--assay exome` or `--assay amplicon` (Table 4): an Agilent SureSelect v5 exome of HG002 (HiSeq 2×125 bp), an Illumina TruSeq exome of HG001 and an Ion AmpliSeq exome of HG002 (Ion Proton, single-end, run with `--platform iontorrent`). Reads over both genes ±10 kb were taken from the producers' hg19 alignments; vendor target intervals were moved to GRCh38 with one offset per locus, and for the Agilent kit, whose BED is not public, the MANE exons ±50 bp served as targets. Truth was GIAB v4.2.1 inside the targets and the high-confidence regions. Inside the targets 33 of 37 SNVs were found. All missed SNVs and indels of the two exomes lie in the 3′ UTR of *BRCA2* exon 27, where the source alignments have 0 to 2 reads; the report lists this stretch as an uncertain interval. The three SNV false positives of the Ion data had allele fractions of 10 to 20 % and carried the allele-balance flag; its 98 false indels are homopolymer errors of the platform, and with `--platform`

`iontorrent` every indel carries a warning. Target QC reported the expected spread: uniformity from 83.9 to 91.6 % and the 1.27 % error rate of the Ion run as a flag.

Table 4. Exome and amplicon data against GIAB v4.2.1, inside the targets. Uniformity: share of target bases with at least 0.2 × the mean depth. Fold-80: mean depth divided by the 20th percentile. Mosaic: candidates reported with `--mosaic`.

Data set	Mean depth	Uniformity	Fold-80	SNVs	SNV FP	Indels	Indel FP	Mosaic
HG002, Agilent SureSelect v5 exome	216×	83.9 %	3.43	10 / 12	0	0 / 1	0	0
HG001, Illumina TruSeq exome	91×	88.7 %	2.22	15 / 17	0	1 / 2	1	0
HG002, Ion AmpliSeq exome (single-end)	200×	91.6 %	2.53	8 / 8	3	0 / 0	98	0

3.6 Low allele fractions

With `--mosaic` the pipeline runs GATK Mutect2 [12] in tumour-only mode with FilterMutectCalls over the same regions and reports PASS calls with an allele fraction of 2 to 30 %, at least three supporting reads and at least 20× depth that the germline caller did not report, as mosaic candidates. The 94 small variants of Section 3.3 were spiked into the real HG002 reads at about 30× with 5, 10, 15, 20 and 30 % of the read pairs drawn from the variant haplotype (Table 5, Figure 7). At 20 % and above every variant was reported, by the germline caller or as a mosaic candidate; at 10 % 65 of 94 and at 5 % 28 of 94. At 30× a 5 % variant is carried by about 1.5 reads on average, so low fractions need the depth of panels or exomes. The negative control and all spiked samples gave 0 false positives. The candidate filters were set after a first run on the exome and amplicon data of Section 3.5, where they removed four TruSeq candidates at 8× with two supporting reads each and 23 Ion indels; that is therefore not an independent test.

Table 5. Spiked variants at low allele fractions in real HG002 reads (about 30×). Mosaic: reported as mosaic candidate; germline: called by HaplotypeCaller (with the allele-balance flag below 25 %).

Fraction	Mosaic	Germline	Either	False positives
5 %	16	12	28 / 94	0
10 %	19	46	65 / 94	0
15 %	15	70	85 / 94	0
20 %	4	90	94 / 94	0
30 %	2	92	94 / 94	0

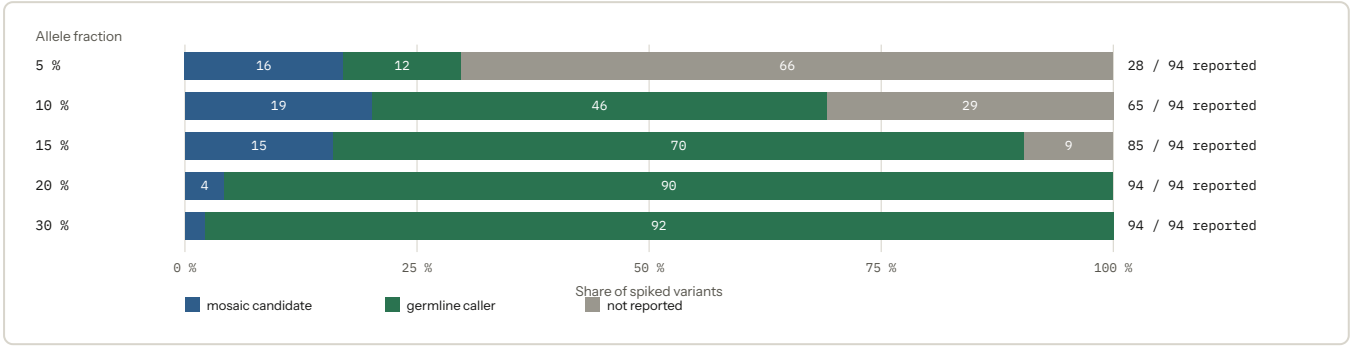


Figure 7. Spiked variants by allele fraction. Each bar is one sample with 94 spiked small variants: reported as mosaic candidate (blue), called by the germline caller (green) or not reported (grey).

3.7 Assembly-based truth

The 39 individuals of the Human Pangenome Reference Consortium [38] that are also part of the 1000 Genomes 30× data [32] provide a truth that does not depend on short reads: the haplotype-resolved long-read assemblies, here as the GRCh38-based VCF of the Minigraph-Cactus pangenome [39]. It covers both genes completely, introns and repeats included. Read pairs were prepared as in Section 3.4 and compared per sample with `vcfeval`; positions where an assembly has no allele were excluded. In the exons ±20 bp 590 of 590 SNVs were found without a false positive (Table 6). 16 of the 18 missed and 9 of the 10 false exon indels lie in one poly-T stretch of the *BRCA1* 3′ UTR (chr17:43,044,804–43,044,817), where assemblies and short reads disagree on the length of the homopolymer; outside it the exon indels were 41 of 43 with one false positive. Over the whole gene loci the figures include every intronic repeat and were not curated.

Variants of 50 bp or more formed three sites. The common intronic Alu insertion in *BRCA2* showed the read-end signature in 6 of 6 carriers and in 0 of 33 non-carriers, while Manta called it in 0. A VNTR in *BRCA1* intron 6 with alleles from −80 to +113 bp was called in 8 of 14 carriers, with 5 calls in 25 non-carriers. A 126 bp insertion that adds

one copy of an adjacent tandem repeat was called in 0 of 19 carriers: the extra copy aligns to the existing one with soft-clipped ends only. The tandem-copy signature (Section 2.2) was written after this result and found the insertion in 19 of 19 carriers and 0 of 20 non-carriers; on the samples of Sections 3.2 and 3.4 that have long reads (Section 3.8) it found 6 of 6 long-read carriers and 0 of 5 non-carriers. Non-carriers had at most one read with the marker base. The thresholds were set after inspecting both sets, so this is not an independent test. Across all 72 samples the only other signature was a 73 bp copy at the VNTR in a sample whose long reads show no insertion there.

As an independent test, 120 further samples of the 1KG-ONT cohort that had not been looked at before were evaluated with the fixed rules on the original 1000 Genomes alignments: 67 of 67 long-read carriers and 1 of 53 non-carriers were reported. Clipped reads alone would have found 60 of the carriers; the marker base and the haplotype recover the rest. The one non-carrier with a report (HG02687) showed all three signals; its long-read genotype rests on 8 spanning reads, 1 of them with the insertion, so it may be a carrier missed by the long reads. The haplotype count agreed with the long reads: all 18 of 18 carriers with two haplotype copies showed the insertion in every spanning read, and 48 of 49 with one copy in only part of them.

Because the rules above were developed on these samples, the final method was frozen and its test registered in the repository before any data were fetched: 200 further 1KG-ONT samples, drawn at random with a fixed seed from the 834 not used before, with the 1000 Genomes alignments as input and the long-read genotypes of Section 3.8 as truth. The 126 bp copy was found in 116 of 116 long-read carriers and reported in 4 of 84 non-carriers; read-end signatures alone found 97 of 116. The *BRCA2* Alu insertion was found in 25 of 25 carriers with no call in 175 non-carriers (read-end signature alone: 19 of 25); its genotype agreed with the long reads in 25 of 25, and that of the tandem copy in 113 of 116. All 4 non-carriers with a call had several concordant signals and only 6, 3, 4, 3 spanning long reads when genotyped again; 3 of them had at least one long read with the insertion, so they may be gaps in the truth rather than false calls; the frozen analysis counts them as false. A registered addendum then added the 980 bp *BRCA1* intronic deletion (1,081 bp in the 1000 Genomes panel) to the catalogue, with breakpoints from the long reads: it was found in 10 of 10 long-read carriers (the 200 samples and all further carriers with 30× data), with no call in 198 non-carriers and the right genotype in 10 of 10. On the same reads, Paragraph [20] given the same allele sequences found 114 of 116 tandem-copy carriers (102 with PASS) and 23 of 25 Alu carriers (12 with PASS), and it called the same 4 long-read non-carriers, which supports that these carry the copy. ExpansionHunter [40] with a locus definition for the tandem array found the copy in 14 of 19 HPRC carriers and 54 of 64 carriers of the independent set.

Table 6. 39 HPRC samples against assembly-based truth, small variants below 50 bp.

Region	Type	Truth	TP	FP	FN	Recall
Exons ±20 bp	SNV	590	590	0	0	100.0 %
Exons ±20 bp	Indel	86	68	10	18	79.1 %
Gene loci	SNV	8,095	8,026	100	69	99.1 %
Gene loci	Indel	3,312	2,494	392	818	75.3 %

3.8 Long reads

The structural variants of Section 3.4 come from short-read callsets. For the 11 of their samples that are in the 1KG-ONT cohort [41], every site was genotyped in the Oxford Nanopore reads: a read spanning the site supports the variant when its alignment carries an insertion or deletion of at least half the expected length there. The pipeline result agreed with the long reads in 26 of 26 sample-site pairs. The long reads disagreed with the panel only for HG01796 at HGSV_216996 (absent in all 26 spanning long reads, carrier in the panel). This is the carrier of the 271 bp deletion without a call in Section 3.4, whose short-read depth showed no deletion either: the panel genotype is most likely wrong.

In addition, all insertions and deletions of 50 bp or more inside both gene loci were collected from the long reads of all 1,019 samples of the cohort and every sample was genotyped at every site (18 sites, none in an exon ±20 bp; 9 seen in a single sample). The intronic *BRCA2* Alu insertion was present in 136 of 1,002 evaluable samples, the 1,081 bp *BRCA1* deletion in 13 of 993 and the 126 bp tandem-repeat insertion of Section 3.7 in 552 of 1,004. The last one is a common allele, which short-read callers miss and only the tandem-copy signature reports (Section 3.7).

3.9 Limits of this validation

- Reads for Sections 3.1, 3.2, 3.4, 3.5 and 3.7 were taken from existing alignments near the genes and realigned against chr13 and chr17 only. This removes competition from paralogous sequence elsewhere, which may overstate precision. A whole-genome run against the complete reference is the next step.
- Spiked variants are synthetic. They test whether a variant that is present in the reads is called and reported, not how real pathogenic rearrangements

look. The spiked Alu element was copied from the reduced reference, which makes its detection easier than a real mobile element insertion.

- No Ensembl VEP cache was available on the validation machine, so annotation and the ACMG/AMP hints were not part of these runs. In Sections 3.1, 3.2 and 3.4 CNVkit was replaced by a neutral stand-in.
- Numbers are small. Ten carriers and twelve large events give wide confidence intervals (Figure 3). The mosaic filters were tuned on the data of Section 3.5.
- The read-end signature was written after the missed Alu insertions had been inspected, so its result on these carriers is not an independent test.
- No real sample with a pathogenic exon deletion, duplication or Alu insertion was tested.

4 Comparison with existing workflows

Table 7. Feature comparison with BRCA-analyzer [5], as described in its repository.

	brca12-wgs 1.1	BRCA-analyzer
Data type	WGS, exome, hybrid-capture and amplicon panels; paired or single-end; Illumina or Ion Torrent	Amplicon panels (MiSeq)
Input	FASTQ, BAM or CRAM	FASTQ
Assembly	GRCh38, enforced; GRCh37 position for every small variant	hg19
Primer removal	samtools ampliconclip, share of clipped reads	cutPrimers, share of trimmed reads
Small variants	GATK 4 HaplotypeCaller or DeepVariant	Pisces, GATK 3 UnifiedGenotyper or FreeBayes
Low allele fraction	Mutect2 mosaic candidates, tested by spike-ins	Pisces calls low fractions
Copy number / SV	CNVkit and exon dosage / Manta, Alu and tandem-copy read-end signatures	Not included
Annotation	VEP, MANE Select, gnomAD, user ClinVar; BIC names	ANNOVAR (licence), snpEff, bundled ClinVar and BIC
Interpretation support	ACMG/AMP criterion hints (ENIGMA v1.2), no classification	Class from ClinVar/BIC
QC	Read QC, per exon and per target coverage, uniformity, fold-80, intervals below 10×	Read counts, amplicon coverage and uniformity
Batches	Sample sheet, cohort table, contamination hint	Joint table, contamination hint
Output	HTML, Markdown, PDF, Excel, JSON; English, German, Russian	Excel, Word; English, Russian
Reproducibility	Nextflow, pinned conda, Docker/Singularity, command log, tests in CI	Python script, Docker image
Open validation	GIAB WGS, exomes and amplicons, real carriers, spike-ins, HPRC assemblies, long reads	Example data
Clinical validation	None (research use only)	More than 900 patient samples with Sanger confirmation, according to its documentation

Licence	MIT	GPL-3.0

5 Limitations

- At 30× WGS, variants below about 15 % allele fraction are often missed even with `--mosaic` (Section 3.6); low fractions need deeper targeted data. Sensitivity is also limited for complex rearrangements and in regions with pseudogene homology such as *BRCA1* exons 1–2.
- Extra copies of a tandem repeat unit are reported only as hints. For the common 126 bp copy (Section 3.7) the thresholds were set on HPRC and ONT samples and then tested on 120 further ONT samples; the haplotype signal relies on linkage that may be weaker in populations not represented in these cohorts. Known alleles are genotyped by realignment only where the allele sequence is in the catalogue (three alleles so far); other tandem copies rely on clipped reads alone and can be missed at low depth, and length alleles of the *BRCA1* intron 6 VNTR would need a repeat-aware genotyper such as ExpansionHunter [40] or danbing-tk [42]. Indel lengths in long homopolymers can differ from assembly-based truth.
- Ion Torrent indels are dominated by homopolymer errors and need confirmation; exons without a target or without reads in targeted data are reported as not assessable, not as negative.
- Copy-number results come from a single sample with a flat reference and are candidates only. In the spike-in experiment CNVkit alone missed every event below 11 kb; detection of exon-level events currently rests on Manta, which needs breakpoints inside mappable sequence.
- Mobile element insertions have no dedicated caller. Manta missed real Alu insertions in several carriers; the read-end signature recognises Alu sequence only, was developed on the same two insertion sites it is reported for here, and has not been tested on an independent set or on L1 and SVA elements (Section 3.4).
- The ACMG/AMP hints cover a subset of criteria. The PVS1 decision tree, functional (PS3/BS3), segregation (PP1), case-control (PS4) and same-codon (PS1/PM5) evidence are not evaluated.
- The pipeline has not been validated for clinical use.

6 Availability and requirements

Repository: https://github.com/pjlabsdev/brca12-wgs · **Archived version:** DOI 10.5281/zenodo.24173638 (version 1.0.1) · **Operating systems:** Linux, macOS · **Requirements:** Nextflow ≥ 23.10, Java 17, conda/mamba, Docker, Singularity or the listed tools, GRCh38 FASTA with index and dictionary, Ensembl VEP GRCh38 cache, ClinVar VCF · **Licence:** MIT.

```
nextflow run main.nf -profile conda \
  --sample S --reads_r1 S_R1.fastq.gz --reads_r2
S_R2.fastq.gz \
  --reference GRCh38.fa --vep_cache ~/.vep --
clinvar_vcf clinvar.vcf.gz
```


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Conflict of interest

None declared.

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